Computer Programming (CSN11101)
ASSIGNMENT-6

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Q1. Write a C program that interchanges the odd and even positions elements in the array. (Hint : if the array size is odd, the last element is not swapped)
A1. CODE:

```
#include <stdio.h>

int main() {
    int n;
    printf("Enter number of elements: ");
    scanf("%d",&n);
    int arr[n];
    printf("Enter elements: ");
    for (int i=0;i<n;i++) scanf("%d",&arr[i]);
    for (int i=0;i<n-1;i+=2){
        int temp=arr[i];
        arr[i]=arr[i+1];
        arr[i+1]=temp;
    }
    printf("New Array : ");
    for (int i=0;i<n;i++) printf("%d ",arr[i]);
    printf("\n");
    return 0;
}
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/Computer programming/Assignment-6$ ./p1
Enter number of elements: 4
Enter elements: 1 4 3 2
New Array : 4 1 2 3
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/Computer programming/Assignment-6$ ./p1
Enter number of elements: 5
Enter elements: 3 5 7 1 3
New Array : 5 3 1 7 3
```

Q2. Write a C program to find min and max elements in an array.
A2. CODE:

```
#include <stdio.h>
int main(){
    int n;
    printf("Enter number of elements: ");
    scanf("%d",&n);
    int arr[n];
    printf("Enter elements: ");
    for(int i=0;i<n;i++) scanf("%d",&arr[i]);
    int mi=arr[0],ma=arr[0];
    for(int i=1;i<n;i++){
        if(arr[i]<mi) mi=arr[i];
        if(arr[i]>ma) ma=arr[i];
    }
    printf("Minimum = %d\n",mi);
    printf("Maximum = %d\n",ma);
    return 0;
}
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/Computer programming/Assignment-6$ ./p2
Enter number of elements: 5
Enter elements: 4 2 9 6 1
Minimum = 1
Maximum = 9
```

Q3. Write a program that initializes an array with values and calculates the sum and average of the elements.
A3. CODE:

```
#include <stdio.h>
int main(){
    int n;
    printf("Enter number of elements: ");
    scanf("%d",&n);
    int arr[n],s=0;
    printf("Enter elements: ");
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
        s+=arr[i];
    }
    float avg=(float)s/n;
    printf("sum=%d\n",s);
    printf("Average=%f\n",avg);
    return 0;
}
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/Computer programming/Assignment-6$ ./p3
Enter number of elements: 5
Enter elements: 4 2 9 1 6
sum=22
Average=4.400000
```

Q4.Write a C program to move all the even numbers to beginning and ODD numbers to the end of the same array.

A4. CODE :

```
#include <stdio.h>
int main(){
    int n;
    printf("Enter number of elements: ");
    scanf("%d",&n);
    int arr[n];
    printf("Enter elements: ");
    for(int i=0;i<n;i++) scanf("%d",&arr[i]);
    int new[n],j=0;
    for(int i=0;i<n;i++) if(arr[i]%2==0){new[j]=arr[i];j++;}
    for(int i=0;i<n;i++) if(arr[i]%2!=0){new[j]=arr[i];j++;}
    printf("New Array: ");
    for(int i=0;i<n;i++) printf("%d ",new[i]);
    printf("\n");
    return 0;
}
```

OUTPUT :

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/Computer programming/Assignment-6$ ./p4
Enter number of elements: 5
Enter elements: 2 5 1 4 3
New Array: 2 4 5 1 3
```

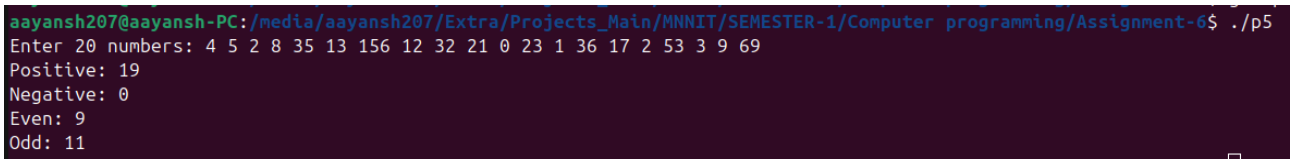
Q5. Twenty numbers are entered through the keyboard into an array. Write a C program to findout how many of them are positive, how many are negative, how many are even and how many are odd using conditional operators.

A5. CODE :

```
#include <stdio.h>
int main(){
    int arr[20],p=0,n=0,e=0,o=0;
    printf("Enter 20 numbers: ");
    for(int i=0;i<20;i++) scanf("%d",&arr[i]);
    for(int i=0;i<20;i++){
        (arr[i]>0)?p++:(arr[i]<0)?n++:0;
        (arr[i]%2==0)?e++:o++;
    }
    printf("Positive: %d\n",p);
    printf("Negative: %d\n",n);
    printf("Even: %d\n",e);
    printf("Odd: %d\n",o);

    return 0;
}
```

OUTPUT :



Q6.Write a C program to find the first peak element of the array which is not smaller than its neighbors.

A6. CODE:

```
#include <stdio.h>
int main(){
    int n;
    int arr[n];
    printf("Enter number of elements: ");
    scanf("%d",&n);
    printf("Enter elements: ");
    for(int i=0;i<n;i++) scanf("%d",&arr[i]);
    int answer=-1;
    for(int i=0;i<n;i++){
        if((i==0||arr[i]>=arr[i-1])&&(i==n-1||arr[i]>=arr[i+1])){
            answer=arr[i];
            break;}
    }
    if(answer==-1){printf("No peak element");}
    else{printf("answer: %d\n",answer);}
}
```

OUTPUT:

```
aayansh207@aayansh-PC:/media/aayansh207/Extra/Projects_Main/MNNIT/SEMESTER-1/Computer programming/Assignment-6$ ./p6
Enter number of elements: 5
Enter elements: 3 4 6 4 7
answer: 6
```